

Test Plan for app.vwo.com

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1. Objective

This document outlines the test plan for **app.vwo.com**, an A/B testing platform that helps websites optimize conversions. The objective is to ensure that all features and functionalities of the platform are thoroughly tested, and work as expected for target users. The goal is to provide a seamless user experience, high reliability, and scalability for clients conducting A/B tests.

2. Scope

Features to be Tested

- **Login Page**
- **Dashboard Page**
- **Create Account Page**
- **AB Testing Campaigns (Creating, Managing, and Analyzing)**
- **User Role Management**

Types of Testing

- **Manual Testing**
- **Automated Testing**
- **Performance Testing (Load & Stress)**
- **Security Testing**
- **Accessibility Testing (WCAG Compliance)**

Test Environments

- **Operating Systems:** Windows 10, macOS, Linux
 - **Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge, Safari
 - **Devices:** Desktop computers, laptops, tablets, smartphones (Android, iOS)
 - **Network Connectivity:** Wi-Fi, cellular, wired
 - **Hardware/Software:** Latest browser versions, system requirements for app operation
 - **Security Protocols:** HTTPS, OAuth tokens, encryption
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3. Inclusions

Test Objectives

- Validate that all features function as per requirements.
- Identify defects in functionality, performance, and security.
- Ensure the platform is scalable and performs under high traffic.
- Validate accessibility for users with disabilities.
- Achieve a certain level of test coverage through both manual and automated testing.

Test Coverage

- Code coverage: Automated tests to achieve at least 80% functional coverage.
 - Functional coverage: All critical user flows will be covered in manual and automated test cases.
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4. Exclusions

- Testing of third-party integrations (if not relevant for the release).
 - Features under active development or in beta will not be tested.
 - Non-critical internal tools (admin panel, reporting system) may be excluded from this phase.
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5. Test Environments

- **Operating Systems:** Windows 10, macOS, Linux
 - **Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge, Safari
 - **Devices:** Desktop computers, laptops, tablets, smartphones
 - **Network Connectivity:** Wi-Fi, Cellular, Wired connections
 - **Test Data Management:** Use mock data or data that simulates real-world user behavior. Reset test data before each cycle to avoid contamination.
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6. Defect Reporting Procedure

Defect Identification Criteria:

- Deviation from functional or non-functional requirements.
- Issues affecting the user experience, security vulnerabilities, or system crashes.

Reporting Process:

- **Tools:** JIRA for defect tracking.
- **Required Information:** Reproduction steps, screenshots/logs, and severity.
- **Triage:** Defects will be categorized by severity (Critical, High, Medium, Low).

- **Roles:** Testers will report defects, developers will fix them, and the Test Lead will oversee the process.
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7. Test Strategy

Step 1: Test Case Creation

- **Techniques:** Equivalence Class Partitioning, Boundary Value Analysis, Decision Table Testing, Use Case Testing.
- **Additional Methods:** Exploratory Testing, Error Guessing.

Step 2: Test Procedure

- **Smoke Testing:** Validate critical features to ensure stability.
- **In-depth Testing:** Execute comprehensive test cases after successful smoke testing.
- **Regression Testing:** Ensure that previously working features are unaffected by new code.
- **Automation:** Automated tests will be executed as part of the CI/CD pipeline.

Step 3: Test Automation Framework

- **Tools:** Selenium, Cypress, Jenkins (for CI/CD integration)
- **Approach:** Automated tests will cover functional testing, regression testing, and cross-browser compatibility.
- **Test Maintenance:** Automated scripts will be updated as per the evolving app features.

Step 4: Performance Testing

- Load testing using JMeter or Gatling to simulate high traffic conditions.
- Stress testing to determine the app's breaking point.

Step 5: Exploratory Testing

- **Focus:** Unscripted testing to explore new or complex features.
 - **Session-Based Testing:** 2-hour exploratory sessions focusing on critical flows.
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8. Test Schedule

Task	Duration Dates	
Test Plan Creation	3 days	May 1–May 3, 2025
Test Case Creation	7 days	May 4–May 10, 2025
Test Execution (Manual)	10 days	May 11–May 20, 2025
Automated Test Creation	7 days	May 11–May 17, 2025

Task	Duration	Dates
Test Execution (Automated)	Ongoing	May 18–May 25, 2025
Test Closure & Reporting	3 days	May 26–May 28, 2025

9. Test Deliverables

- **Test Plan:** Detailing scope, objectives, and testing approaches.
 - **Test Cases:** Comprehensive functional and non-functional test cases.
 - **Defect Reports:** Detailed defect reports for issues found.
 - **Test Execution Results:** Pass/fail status of executed tests.
 - **Test Summary Report:** High-level summary of testing outcomes.
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10. Entry and Exit Criteria

Entry Criteria:

- Test cases and scenarios are created and signed off.
- Application is stable and ready for testing.

Exit Criteria:

- All critical defects are resolved.
 - Test case execution is complete.
 - Test reports are generated, and results shared with stakeholders.
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11. Tools

- **JIRA:** Bug tracking and defect management.
 - **Selenium:** Automated functional testing.
 - **Jenkins:** Continuous Integration for automated test execution.
 - **JMeter/Gatling:** Load and performance testing.
 - **Axe:** Accessibility testing.
 - **Postman:** API testing (if applicable).
 - **Mind Map Tool:** For brainstorming and test scenario design.
 - **Snipping Tool:** For capturing screenshots.
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12. Risks and Mitigations

Possible Risks:

- **Non-Availability of Resources:** Mitigation: Backup resource planning and cross-training team members.
 - **Build URL Not Accessible:** Mitigation: Maintain alternate test environments and work on other tasks if the build is down.
 - **Limited Testing Time:** Mitigation: Prioritize critical features, extend test execution hours.
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13. Approvals

- **Documents for Client Approval:**
 - Test Plan
 - Test Scenarios
 - Test Cases
 - Test Execution and Defect Reports
 - Test Summary Report

Approval Workflow:

- **Test Plan Approval:** Sign-off by stakeholders before testing starts.
 - **Test Execution Reports:** Regular updates to stakeholders.
 - **Final Test Report:** Client and management sign-off after completion of testing.
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Additional Sections

Test Case Traceability Matrix

A matrix mapping **functional requirements** to corresponding **test cases** to ensure full coverage.

Requirement ID Feature		Test Case ID	Test Case Description
REQ-001	Login Page	TC-001	Verify login with valid credentials
REQ-002	Dashboard Page	TC-002	Verify dashboard displays A/B testing results

Security Testing

- **Penetration Testing:** Conduct penetration tests to identify vulnerabilities.
- **Vulnerability Scanning:** Use tools like **OWASP ZAP** to identify common vulnerabilities.

Accessibility Testing

- **WCAG Compliance:** Use **axe** and manual testing to ensure compliance with WCAG 2.1 standards.

Conclusion

This test plan provides a structured and comprehensive approach to testing **app.vwo.com**. It ensures all critical aspects of the application—such as functionality, performance, security, and accessibility—are tested thoroughly. By integrating manual and automated testing, performance testing, and robust defect management procedures, this plan ensures high-quality deliverables and a smooth user experience.